

Ravi Shyam Singh

Software Engineer

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Education

Kalasalingam Academy of Research and Education

Srivilliputhur, TN

B.Tech in Computer Science and Engineering — CGPA: 7.6/10.0

May 2026

- **Coursework:** Java, Software Engineering, Operating Systems, Algorithms, Artificial Intelligence, Database Management Systems

Skills & Technologies

Languages: Java, TypeScript, Python, C++, SQL

Databases: PostgreSQL, MySQL, pgvector, Redis

Frontend: React, Vite, Tailwind CSS, HTML5, CSS3, Responsive Design

Backend: Node.js, Express.js, Flask, FastAPI, Django, REST APIs, Role-Based Access Control (RBAC), Socket.IO

AI & Tools: LangGraph, Retrieval-Augmented Generation (RAG), Hugging Face Inference API, Gemini API, SentenceTransformers, Embeddings, Vector Search, HNSW Indexing

Cloud, Infrastructure & DevOps: Nutanix Architecture, Microservices Architecture, Firebase Authentication, Docker, Git, GitHub, Render, Vercel, CI/CD, Database Migrations

Engineering Practices: Agile Development, Code Review, API Documentation, Async Programming, Event-Driven Architecture, Error Handling, Audit Logging, Unit Testing, System Design

Projects

SyncSpace — Multi-Agent AI Workspace | *React, Node.js, FastAPI, PostgreSQL, LangGraph, RAG*

[GitHub](#) · [Live](#)

- Built a full-stack AI workspace that converts manager-created tasks into live, multi-agent workflows and structured downloadable reports.
- Developed React-based dashboards with Firebase authentication, role-based access control, and Socket.IO realtime task/subtask monitoring.
- Implemented Node.js/Express APIs with PostgreSQL for task, report, audit-log, and user lifecycle management.
- Created a FastAPI + LangGraph orchestration service using RAG, pgvector, Hugging Face inference, parallel specialist agents, and durable workflow checkpoints.
- Built a secure Admin Panel for user/role management, task moderation and retries, report control, service monitoring, and audit-log tracking.

Interactive Java Learning Platform | *React, Firebase, LangGraph, Gemini API*

[GitHub](#) · [Live](#)

- Built a full-stack Java learning platform used by 200+ students, delivering 99% accurate content behind a 99% secure login system.
- Implemented Firebase authentication, protected routes, and real-time sync, enabling 100% course completion tracking.
- Built an AI-powered quiz generation module with LangGraph and Gemini API for module-wise dynamic question creation; debugged real-time sync issues across Firebase and React Query.

Smart Gym Access & Management Platform | *React, Node.js, Express, MongoDB*

[GitHub](#) · [Live](#)

- Engineered browser-based facial recognition with face-api.js and TensorFlow.js for zero-contact member verification at 95%+ accuracy, eliminating hardware costs entirely.
- Implemented Firebase Auth with role-based access control and automated attendance tracking, cutting manual admin overhead by 80%.
- Built a full-stack, PWA-style app with React, Tailwind CSS, Node.js, Express, and MongoDB; debugged face-api.js recognition edge cases across browsers.

Blood Bank Management System (BDMS) | *Django, SQL*

[GitHub](#)

- Engineered a Django platform serving 100+ users with role-based access and intuitive dashboards, cutting user training time by 25%.
- Automated donor/request alerts via SMTP, improving response time by 50% and accuracy by 20%.
- Wrote unit tests for role-based access logic, reducing post-deployment bugs by 30%.

Certifications

Full Stack Developer Bootcamp — Master Frontend to Backend, GeeksforGeeks, 2024 — [Link](#)

Solutions Architecture Job Simulation, AWS, 2024 — [Link](#)

5-Day AI Agents Intensive Course, Google, 2025 — [Link](#)

Google for Startups: Prompt to Prototype, Google, 2025 — [Link](#)

Activities

State-Level Hackathon on Website Development — Top 7 of 700+ participants, A.K.D Dharmaraja College — [Link](#)

TechUstav Synergicz 2.0 — Top 5 of 450+ participants, Thiagarajar College of Engineering — [Link](#)

Campus Ambassador, BECon '26 (EDC IIT Delhi) — Outreach & engagement initiatives, Dec 2025–Feb 2026 — [Link](#)

IEEE ICCiEST 2026 — Accepted research paper on Blood Bank Donor Management System — [Link](#)

Languages

Hindi (Native) · English (Fluent) · Tamil (Basic) · Bengali (Basic)